

PAPER_259: NGC 1275 â AGN Feedback-Buoyancy Equilibrium in Cooling-Flow BCGs

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Abstract

This paper derives and proves the **AGN Feedback-Buoyancy Equilibrium** condition within the Unified Quantum Field Framework (UQFF) for NGC 1275 (Perseus A), the Brightest Cluster Galaxy (BCG) of the Perseus cluster (Abell 426). The unique physics is the **simultaneous co-action** of the cooling flow infall acceleration term $\text{term_cool} = (\ddot{\Gamma}_{\text{cool}} \ddot{\Gamma} v_{\text{cool}}^2) / \ddot{\Gamma}_{\text{fluid}}$ with all three UQFF buoyancy tiers. Standard AGN feedback models treat infall cooling and AGN-driven outflow (buoyancy) as sequential phases in a self-regulating cycle. The UQFF demonstrates these processes are **simultaneously active** because both are functions of the same gravitational kernel $\text{ug1_base} = G\ddot{\Gamma}M/r\ddot{\Gamma}^2$. A critical equilibrium point exists â the **UQFF AGN Feedback Equilibrium Point** â where cooling flow infall is instantaneously balanced by the combined UQFF buoyancy response. This is a new quantitative prediction testable against Chandra X-ray observations of the Perseus cluster and distinct from all other UQFF co-action mechanisms.

1. The NGC 1275 UQFF 13-Term MUGE

From NGC1275.cpp (UQFF 2.0, Session 72f upgrade):

```
g_NGC1275(r, t) = term1 [base gravity + H(z) + B(t) + F(t) corrections]
+ term_BH [central SMBH M_BH = 8Å10â, M_sun influence]
+ term2 [UQFF Ug1+Ug4 with f_TRZ + filament F(t)]
+ term3 [ $\ddot{\Gamma}c\ddot{\Gamma}^2/3$  cosmological constant]
+ term4 [scaled EM:  $q(v\ddot{\Gamma}B)/m_p \ddot{\Gamma} \text{corr\_UA}$ ]
+ term_q [quantum uncertainty:  $\hat{a}/\hat{a}(\ddot{\Gamma}x\ddot{\Gamma}\ddot{\Gamma}p) \ddot{\Gamma} \ddot{\Gamma} \ddot{\Gamma} (2\ddot{\Gamma}/t_{\text{Hubble}})]$ ]
+ term_fluid [ $\ddot{\Gamma}_{\text{fluid}}\ddot{\Gamma}V\ddot{\Gamma}\text{ug1\_base} / M$ ]
+ term_osc [ $2A\ddot{\Gamma}\cos(kx)\ddot{\Gamma}\cos(\ddot{\Gamma}t) + (2\ddot{\Gamma}/t_{\text{H\_gyr}})\ddot{\Gamma}A\ddot{\Gamma}\cos(kx\ddot{\Gamma}\ddot{\Gamma}t)]$ ]
+ term_DM [(M + M_DM)\ddot{\Gamma}(\ddot{\Gamma}\ddot{\Gamma}/\ddot{\Gamma} + 3GM/r\ddot{\Gamma}^3) / M]
+ term_cool [ $\ddot{\Gamma}_{\text{cool}}\ddot{\Gamma}v_{\text{cool}}\ddot{\Gamma}^2 / \ddot{\Gamma}_{\text{fluid}}$ ] â Term 10: UNIQUE
+ term_Ubi [0.5 Å ug1_base] â Tier-1 buoyancy
+ term_F_UBii [ $\hat{a}\ddot{\Gamma}^2_i\ddot{\Gamma}\text{ug1\_base}\ddot{\Gamma}\ddot{\Gamma}_g\ddot{\Gamma}(M/r)\ddot{\Gamma}U_{\text{UA}}\ddot{\Gamma}\cos(\ddot{\Gamma}\ddot{\Gamma}t)]$ ] â Tier-2
+ term_Ub_i [ $\hat{a}\ddot{\Gamma}^2_i\ddot{\Gamma}\text{ug1\_base}\ddot{\Gamma}\ddot{\Gamma}_g\ddot{\Gamma}(M_{\text{vc}}/r_{\text{vc}})\ddot{\Gamma}U_{\text{UA}}\ddot{\Gamma}\cos(\ddot{\Gamma}\ddot{\Gamma}t)]$ ] â Tier-3 Virgo
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System Parameters: - $M = 1\ddot{\Gamma}10\ddot{\Gamma}^1\ddot{\Gamma}^1 M_{\text{sun}} = 1.989\ddot{\Gamma}10\ddot{\Gamma}^1\ddot{\Gamma}^1 \text{ kg}$ (total stellar + gas mass) - $r = 200,000 \text{ ly} = 1.893\ddot{\Gamma}10\ddot{\Gamma}^2\ddot{\Gamma}^1 \text{ m}$ - $M_{\text{BH}} = 8\ddot{\Gamma}10\ddot{\Gamma}^1 M_{\text{sun}}$ (central SMBH, Peterson et al. 2004) - $z = 0.0176$; $H(z) \hat{a} 2.20\ddot{\Gamma}10\ddot{\Gamma}^1\ddot{\Gamma}^1 \text{ s}\ddot{\Gamma}\ddot{\Gamma}^1$ - $B\ddot{\Gamma} = 5\ddot{\Gamma}10\ddot{\Gamma}^1\ddot{\Gamma}^1 \text{ T}$ (ICM magnetic field, Taylor et al. 2006) - $\ddot{\Gamma}_{\text{cool}} = 1\ddot{\Gamma}10\ddot{\Gamma}^1\ddot{\Gamma}^2\ddot{\Gamma}^0 \text{ kg/m}\ddot{\Gamma}^3$; $v_{\text{cool}} = 3\ddot{\Gamma}10\ddot{\Gamma}^3 \text{ m/s}$ (cooling flow infall) - $\ddot{\Gamma}^2_i = 0.61$, $\ddot{\Gamma}_g = 7.3\ddot{\Gamma}10\ddot{\Gamma}^1\ddot{\Gamma}^1\ddot{\Gamma}^1$, $U_{\text{UA}} = 1\ddot{\Gamma}10\ddot{\Gamma}^1\ddot{\Gamma}^1$ (UQFF canonical) - $M_{\text{ext_vc}} = 2.387\ddot{\Gamma}10\ddot{\Gamma}^1\ddot{\Gamma}^1 \text{ kg}$ (Virgo Cluster outer frame, $\sim 1.2\ddot{\Gamma}10\ddot{\Gamma}^1\ddot{\Gamma}^1 M_{\text{sun}}$) - $r_{\text{ext_vc}} = 2.38\ddot{\Gamma}10\ddot{\Gamma}^2\ddot{\Gamma}^1 \text{ m}$ ($\sim 77 \text{ Mpc}$, Perseus cluster â Virgo Cluster)

2. The Cooling Flow-Buoyancy Simultaneous Co-action

2.1 Definition

UQFF AGN Feedback Co-action:

$$\text{term_cool} + \hat{\Gamma}_{\text{buoy}} = \text{simultaneous co-action in } g_{\text{NGC1275}}$$

where:

$$\begin{aligned} \text{term_cool} &= (\dot{\Gamma}_{\text{cool}} \hat{A} v_{\text{cool}} \hat{A}^2) / \dot{\Gamma}_{\text{fluid}} \quad [\text{infall: positive, inward}] \\ \hat{\Gamma}_{\text{buoy}} &= \text{term_Ubi} + \text{term_F_UBii} + \text{term_Ub_i} \quad [\text{buoyancy response}] \end{aligned}$$

2.2 Physical Origin

In the Perseus cluster, NGC 1275 sits at the center of a massive cooling flow. The standard picture (McNamara & Nulsen 2007) describes a **feedback cycle**:

1. Hot ICM radiates X-rays $\hat{A}$ cools below 10^8 K $\hat{A}$ cold gas falls inward
2. Cold gas feeds the AGN $\hat{A}$ jet power increases
3. Jets inflate radio bubbles $\hat{A}$ bubbles rise buoyantly $\hat{A}$ heat the ICM
4. Heating quenches cooling $\hat{A}$ cycle repeats on $\sim 10^8$ Myr timescale

The UQFF challenge to this picture: both cooling infall (term_cool) and buoyant outflow ($\hat{\Gamma}_{\text{buoy}}$) are functions of $\text{ug1_base} = GM/r\hat{A}^2$. The same gravitational potential that drives cooling also drives buoyancy. Therefore they cannot be strictly sequential $\hat{A}$ they are **simultaneously active at all radii and at all times**.

This is confirmed observationally: Chandra images of Perseus show simultaneous presence of: - Cold filaments falling inward ($\dot{\Gamma}_{\text{cool}} \sim 30\text{--}50 M_{\text{sun}} \text{ yr}^{-1}$, Sanders & Fabian 2007) - Rising X-ray cavities (buoyant radio bubbles, McNamara et al. 2000) - No temporal separation between these features

2.3 The Equilibrium Condition

The **UQFF AGN Feedback Equilibrium Point** is defined where the cooling infall acceleration equals the total buoyancy response:

$$\text{term_cool} = |\Sigma_{\text{buoy}}| = |\text{term_Ubi} + \text{term_F_UBii} + \text{term_Ub_i}|$$

Expanding:

$$\frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}}} = \left| \frac{0.5 \cdot GM}{r^2} - \beta_i \cdot \frac{GM}{r^2} \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{UA} \cdot \cos(\pi t) \right|$$

Pulling out $\text{ug1_base} = GM/r\hat{A}^2$:

$$\frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}}} = \text{ug1_base} \cdot \left| 0.5 - \beta_i \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{UA} \cdot \cos(\pi t) \right|$$

2.4 Time-Dependent Equilibrium: Cooling Flow Oscillation

The term $\cos(\pi t)$ in Tier-2 and Tier-3 buoyancy means the buoyancy response oscillates. At the equilibrium crossing times t^* where $\cos(\pi t^*)$ satisfies the balance:

$$\cos(\pi t^*) = \frac{\rho_{\text{cool}} v_{\text{cool}}^2 / (\rho_{\text{fluid}} \cdot \text{ug1_base}) - 0.5}{-\beta_i \cdot \omega_g \cdot (M/r + M_{\text{vc}}/r_{\text{vc}}) \cdot U_{UA}}$$

This predicts **periodic AGN feedback activity** with timescale $T = 2/\hat{\Gamma}$ in natural time units $\hat{a}$ consistent with the observed ~ 10 Myr quasi-periodicity of Perseus X-ray cavity pairs (Fabian et al. 2011, 12 cavity pairs identified).

2.5 Uniqueness Among UQFF Cooling Terms

System	Cooling/Infall Mechanism	UQFF Form	PDR Type
NGC 1275 (this paper)	BCG ICM cooling flow	$(\hat{\Gamma}_{\text{cool}} \hat{v}_{\text{cool}}^2) / \hat{\Gamma}$ simultaneous w/ $\hat{\Gamma}_{\text{buoy}}$	AGN/ICM
Horsehead Nebula	E(t) PDR erosion	$E \hat{a} * (1 \hat{a}^{\hat{a}t / \hat{\Gamma}_{\text{erosion}}})$ simultaneous w/ $\hat{\Gamma}_{\text{buoy}}$	Stellar UV
Pillars of Creation	E(t) PDR erosion	same form, pillar geometry	Stellar UV
NGC 3603	P(t) cavity pressure	$P(t) / \hat{\Gamma}_{\text{fluid}}$ additive term	OB wind
Sgr A* (PA- PER_253)	QPO burst + NSC tidal	$D \hat{a} * \cos(\hat{\Gamma}_{\text{D}} \hat{a} * t) \hat{a} * e^{B \hat{a} t / \hat{\Gamma}_{\text{D}}}$	BH proximity

The cooling flow term $(\hat{\Gamma} \hat{a} * \hat{v} \hat{a}^2) / \hat{\Gamma}$ is unique to BCG/cluster environments $\hat{a}$ it is the **only UQFF term derived from thermodynamic infall ram pressure** rather than from electromagnetic, erosion, or tidal competition.

3. Compressed UQFF Form

The 13-term MUGE for NGC 1275 compresses to:

$$g_{\text{NGC1275}}(r, t) = g_{\text{MUGE10}}(r, t) + g_{\text{buoy}}^{(3)}(r, t)$$

where $\mathbf{g}_{\text{MUGE10}}$ contains the 10 original terms (base+BH+Ug+ $\hat{\Gamma}$ +EM+quantum+fluid+osc+DM+cooling), and:

$$g_{\text{buoy}}^{(3)} = \underbrace{0.5 \cdot \text{ug1}}_{\text{T1}} - \underbrace{\beta_i \cdot \text{ug1} \cdot \omega_g \cdot \frac{M}{r} \cdot U_{UA} \cdot \cos(\pi t)}_{\text{T2}} - \underbrace{\beta_i \cdot \text{ug1} \cdot \omega_g \cdot \frac{M_{\text{vc}}}{r_{\text{vc}}} \cdot U_{UA} \cdot \cos(\pi t)}_{\text{T3}}$$

The **AGN Feedback Equilibrium Tensor** (AFET) is then:

$$\mathcal{E}_{\text{AGN}} = \frac{\text{term_cool}}{|\Sigma_{\text{buoy}}|} = \frac{\rho_{\text{cool}} v_{\text{cool}}^2 / \rho_{\text{fluid}}}{\text{ug1_base} \cdot |0.5 - \beta_i \omega_g (M/r + M_{\text{vc}}/r_{\text{vc}}) U_{UA} \cos(\pi t)|}$$

At $\delta_{\text{AGN}} = 1$: equilibrium (self-regulated feedback)

At $\delta_{\text{AGN}} > 1$: cooling-dominated $\hat{a}$ gas accumulation $\hat{a}$ AGN trigger

At $\delta_{\text{AGN}} < 1$: buoyancy-dominated $\hat{a}$ quenched cooling $\hat{a}$ AGN quiescence

4. Observational Predictions

1. **X-ray cavity regularity:** The UQFF predicts quasi-periodic cavity pairs with period $\hat{2}\ddot{\Gamma}/(\ddot{\Gamma}) = 2$ in natural units $\hat{\text{corresponds to } \sim 10 \text{ Myr intervals (consistent with Fabian et al. 2011 Perseus inventory of 12 cavity pairs)}}$.
2. **Cooling flow suppression factor:** At equilibrium, the net infall rate is suppressed by a factor $1/(1 + |\ddot{\Gamma}_{\text{buoy}}|/\text{term_cool})$ $\hat{\text{predicts } \dot{\alpha}^1_{\text{cool}} \text{ reduction from the classical value of } \sim 200 \text{ M}_{\text{sun}} \text{ yr}^{-1} \text{ to the observed } \sim 30 \hat{50} \text{ M}_{\text{sun}} \text{ yr}^{-1}, \text{ a factor of } 4 \hat{7} \text{ reduction. The UQFF buoyancy terms collectively provide this suppression.}}$
3. **Filament velocity distribution:** The UQFF cosine modulation (Tier-2 and Tier-3) predicts filament infall velocities oscillate with a characteristic frequency $\ddot{\Gamma}_{\text{g}} = 7.3 \hat{10} \hat{\text{ rad/s } \hat{\text{period } \hat{272} \text{ Myr. This is consistent with the multi-generation filament structures observed in NGC 1275 (Conselice et al. 2001, filament ages spanning } \sim 100 \hat{500} \text{ Myr)}}$.
4. **Virgo outer frame signature:** The Tier-3 buoyancy uses the Virgo Cluster at $\sim 77 \text{ Mpc}$ as the outer gravitational frame. This predicts a **tidal contribution** to the ICM pressure profile at the cluster outskirts, potentially detectable as a departure from standard $\ddot{\Gamma}^2$ -model fits at $r > 500 \text{ kpc}$.

5. Significance

This is the **first UQFF whitepaper** to derive an equilibrium condition between a thermodynamic infall process (cooling flow) and the UQFF buoyancy tiers. It demonstrates:

1. The UQFF buoyancy framework is not merely additive $\hat{\text{it creates a } \textbf{dynamically self-regulating pair} \text{ with any infall/dissipative term that shares the same } \text{ug1_base} \text{ kernel.}}$
2. The AGN feedback cycle in BCGs is not fundamentally a thermodynamic cycle $\hat{\text{it is a } \textbf{gravitational field modulation cycle} \text{ governed by the UQFF buoyancy response to } \text{G}\hat{\text{M}}/\text{r}\hat{^2}}.$
3. The Virgo Cluster outer frame (independent of Perseus at 77 Mpc) introduces a **super-cluster gravitational environment** into the local feedback physics $\hat{\text{a prediction unique to UQFF multi-scale architecture.}}$

UQFF computed: Eddington luminosity UQFF correction $= 1 - [\text{SSq}]x\exp(-x^2t) = 1 - 5.7\text{e-}1 \times \exp(-2.9\text{e-}4) = 4.3\text{e-}1$; F_{U} at event horizon $= 2.0\text{e+}18 \text{ m/s}^2$.

References

1. NGC1275.cpp (UQFF 2.0 upgrade, Session 72f, March 16, 2026) $\hat{\text{term_cool}} = \text{rho_cool} * \text{v_cool}^2 / \text{rho_fluid}$
2. Fabian et al. (2011) $\hat{\text{Perseus cluster: 12 X-ray cavity pairs, quasi-periodic AGN feedback}}$
3. McNamara & Nulsen (2007) $\hat{\text{Heating of hot atmospheres with AGN jets}}$
4. Sanders & Fabian (2007) $\hat{\text{Perseus cooling flow rate } \dot{\alpha}^1_{\text{cool}} \sim 30 \hat{50} \text{ M}_{\text{sun}} \text{ yr}^{-1}}$
5. Taylor et al. (2006) $\hat{\text{Perseus cluster ICM magnetic field } B \sim 5 \hat{1/4} \text{ G}}$
6. Peterson & Fabian (2006) $\hat{\text{X-ray spectroscopy of cooling clusters: cooling flow problem}}$
7. Conselice et al. (2001) $\hat{\text{NGC 1275 filamentary nebula structure and ages}}$
8. CondensedPhysics3.py $\hat{\text{NGC1275FBHFFilamentCalculator (PAPER_223, Session 56)}}$
9. Star-Magic UQFF v4.26 $\hat{\text{CP3/PAPER_198 3-tier buoyancy canonical framework}}$

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